

Abel Zelalem

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Education

Bachelor of Science in Computer Engineering

Expected May 2027

University of Nevada, Las Vegas (UNLV) | GPA: 3.15

Relevant Coursework: Embedded Systems Design | Digital System Architecture and Design | Digital System Design | Computer Logic Design II | Electronics I | Circuits II | Operating Systems | Data Structures

Associate of Science

May 2024

College of Southern Nevada

Technical Skills

Embedded and Hardware: Microcontrollers | I2C | SPI | UART | PWM | Digital Logic | Analog Circuits | PCB Design (KiCad) | Oscilloscope | LTspice

Languages: C | C++ | SystemVerilog | Verilog | RAPID | Python | Java (Familiar)

Architectures: AVR (ATmega328P/PB, ATmega2560) | ARM | RISC-V | MIPS

Robotics & CAD: ABB RobotStudio | Pick-and-Place Automation | Fusion 360 | 3D Printing

Tools: Git | Microchip Studio | ModelSim | Quartus II | MATLAB

Experience

Robotics Programmer

January 2026 - June 2026

Naama Hair and Wigs LLC

Las Vegas, NV

- Programmed an ABB YuMi IRB 14000 dual-arm collaborative robot in RAPID and RobotStudio as the sole robotics engineer at an early-stage startup, building the full motion control pipeline from scratch for wig-assembly automation.
- Demonstrated the core assembly step end to end: the robot threaded raw hair through wig lace and held it in position through the adhesive cure, producing a bonded result for a task that previously required fully manual work.
- Designed and 3D printed hair-positioning jigs in Fusion 360 and defined fixture requirements with two mechanical engineering students to extend the process toward full-lace assembly.

Store Associate

November 2021 - November 2023

In-N-Out Burger

Las Vegas, NV

- Worked at one of the highest-volume In-N-Out locations in the country, consistently meeting speed and quality targets during peak rushes while coordinating with a team of 15–20 people.
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Project Experience

Microcontroller-Based Quadcopter Drone, Embedded Systems Design

May 2026

- Collaborated on a 3-person team to design and build a custom quadcopter from scratch on an ATmega328PB, integrating I2C-based IMU sensor fusion, PID stabilization, and Bluetooth Xbox controller input over UART, achieving momentary hover under a bench power supply; sustained flight testing was limited by battery availability.
- Designed and 3D printed the drone's custom X-configuration chassis in CAD, then wired and calibrated the ESCs and programmed PWM output for four brushless motors, debugging motor spin direction and calibration issues until all four motors responded identically.

Pulse Width Modulator (PWM) Circuit, Circuits II

August 2025

- Designed and validated a 555-timer-based PWM circuit with diode-separated charge and discharge paths, confirming stable 25, 50, and 75 percent duty cycles against LTspice simulations using oscilloscope measurements.

SystemVerilog Elevator, Digital Logic Design II

April 2025

- Designed a 4-floor elevator control system in SystemVerilog as a finite state machine on an Altera DE2-115 FPGA, handling directional movement, internal and external floor requests, door control, and a 7-segment floor display.
- Built a hardware debounce module to eliminate false button triggers and tuned clock timing for visible transitions, verifying functional correctness and edge cases such as equal current and desired floors in ModelSim.